Tao Wen

Contact

Associate Professor, Department of Earth and Environmental Sciences
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Interests

My research examines water and elemental cycles in coupled human–natural systems, integrating large-scale data science with field and laboratory studies. I direct the Syracuse University **H**ydrogeochemistry **A**nd e**N**vironmental **D**ata **S**ciences (**HANDS**) and **N**oble Gases in **E**arth **S**ystems **T**racing (**NEST**) Research Laboratories. I teach undergraduate and graduate topics in water and the environment, hydrogeology, aqueous geochemistry, Earth and environmental data analysis, and machine learning.

Education

2011-2017 Ph.D., Department of Earth and Environmental Sciences; University of Michigan

2011-2014 M.S., Department of Earth and Environmental Sciences; University of Michigan

2007-2011 B.S., School of Earth and Space Sciences; University of Science and Technology of China

Positions Held

2026-present Associate Professor, Department of Earth and Environmental Sciences, Syracuse University

2020-2026 Assistant Professor, Department of Earth and Environmental Sciences, Syracuse University

2017-2020 Postdoctoral Scholar, Earth and Environmental Systems Institute, Pennsylvania State University

Awards and Honors

2026 40 Under 40 Recognition Program Award, American Academy of Environmental Engineers and Scientists

2025 Emerging Investigator Series Award, International Association of GeoChemistry

2021 Excellence in Review Award from Applied Geochemistry, International Association of GeoChemistry

2019 Office of Postdoctoral Affairs Travel Award, Pennsylvania State University

2012-2016 Rackham Conference Travel Award, University of Michigan

2016 Best Student Poster Presentation, American Institute of Professional Geologists Michigan Section

2015-2016 Scott Turner Research Awards, University of Michigan

2012 Stewart R. Wallace Fellowship, University of Michigan

2011 Excellent Undergraduate Researcher Award, University of Science and Technology of China

2010 Guanghua Education Scholarship, University of Science and Technology of China

2008-2009 Outstanding Student Scholarship, University of Science and Technology of China

Grants

Summary *Total funding awarded as PI: $1,180,599, total funding awarded as co-PI and PI: $8,002,086*

Federal

07/2026 – 03/2027 Physics-Informed AI for Accelerated Discovery of Ion-Adsorption-Type Rare Earth Element Resources in Granitic Systems, Department of Energy (lead institution: Oak Ridge National Laboratory), lead PI: Gu, X., Syracuse University PI: **Wen, T.**, Co-PIs: Lu, D., Chai, C., Zhu, M., Li, L., $134,686 (total grant $750,000), 07/2026 – 03/2027.

09/2026 – 08/2027 Conference: GEO-CM: Workshop: Envisioning the Future of Critical Minerals and Materials Cyberinfrastructure, National Science Foundation EAR-Geoinformatics, lead institution: University of Idaho, PI: Ma, X., Co-PIs: Morrison, S.M., **Wen, T.**, Prabhu, A., $183,252, 09/2026 – 08/2027.

2024-2027 EA: Acquisition of a Triple Quadrupole ICP-MS at the Department of Earth and Environmental Sciences, Syracuse University, National Science Foundation EAR/IF, PI: Lu, Z., Co-PIs: **Wen, T.**, Ivany, L., $209,493.

2024 – 10/2027 Developing a Data-Driven Framework to Assess the Status and Environmental Impacts of Marginal Oil and Gas Wells, Department of Energy NETL, PI: **Wen, T.**, $562,432, through 10/2027.

2022-2026 Collaborative Research: Frameworks: DeCODER (Democratized Cyberinfrastructure for Open Discovery to Enable Research), National Science Foundation CSSI, PI: **Wen, T.**, $460,281 (total grant: $3,200,000).

2020 Developing Online Learning Module: Intelligent Earth Computational and Data Science Methods for Research, National Science Foundation HydroLearn Fellowship, PI: **Wen, T.**, $3,000.

State

08/2026 – 08/2027 SHARCS: Sampling for Healthy Aquatic Resources with Community Support, NYS Center of Excellence in Healthy Water Solutions, PI: Ninokawa, A., Co-PI: **Wen, T.**, $29,709, 08/2026 – 08/2027.

2025-2026 SHARCS: Sampling for Healthy Aquatic Resources with Community Science, NYS Center of Excellence, PI: Ninokawa, A., Co-PIs: **Wen, T.** and others, $30,000.

Other

10/2026 – 10/2029 The RhINO-2 Project: Rift-Climate-Earth Surface Interactions: Rift Basin Stratigraphy and Implications for Global Exploration, Petrobras, PI: Scholz, C., Co-PIs: Mouch, R., **Wen, T.** and Junium, C., $2,500,000, 10/2026 – 10/2029.

2024-2025 Safeguarding Syracuse Communities from Urban Flooding and Water Quality Issues Using Real-time Stream Monitoring Techniques, Syracuse University Engaged Communities Network Mini Grant, PI: **Wen, T.**, $10,000.

2024-2025 Evaluating Long-Term Water Quality Trend in an Urban Stream with Mixed Human-Built Infrastructure, Syracuse University Faculty Creative Activities and Research (FCAR) Grant Program, PI: **Wen, T.**, $3,000.

2022-2026 An Analysis of Water Quality and Exposures in the Oldest Hydrocarbon Basin in the USA, Health Effects Institute Energy, PI: Baka, J., Co-PIs: Brantley, S., **Wen, T.**, $16,599 (total grant $250,000).

2022 Evaluating the Impact of Land Management and Land Cover on the Release of Cemetery-derived Nutrients and Heavy Metals to Surface Water, Syracuse University Office of Undergraduate Research and Creative Engagement (SOURCE), PI: **Wen, T.**, $7,200.

2022 Support for the 2022 Shale Network Workshop, Oak Ridge Associated Universities, PI: Baka, Co-PI: **Wen, T.**, $4,000.

2021-2023 Analyzing Community Exposures to Potential Ground Water Contamination in a Region with Overlapping Energy Extraction Activities in Southwestern Pennsylvania, Health Effects Institute Energy, PI: Baka, J., Co-PIs: Brantley, S., **Wen, T.**, $7,185 (total grant $250,000).

2021-2022 Developing a Cloud-based Open-Source Platform for an Automatic High-throughput Monitoring System to Safeguard Stream Water Quality, Earth Science Information Partners, PI: **Wen, T.**, Co-PIs: Xue, L., Agarwal, A., Woda, J., Castronova, A., Zhang, S., $10,000.

Publications

* = Equal contributions and/or corresponding author
† = Postdoc author, directly supervised by Wen at time of publication
†† = Graduate student author, directly supervised by Wen at time of publication
††† = Undergraduate student author, directly supervised by Wen at time of publication
§ = Student or postdoc author of collaborator at time of publication

Journal Articles

in review §Fenske, A., ††Su, J., Baka, J., Forsythe, B., Williams, L., Shaheen, S., **Wen, T.**, Brantley, S.L. and Warner, N.R., Localized Effects Amid Ong-Term Declines in Radon Near Unconventional Oil and Gas Activity. Environmental Science & Technology

in review §Johnson, K., Sethna, L.R., Atwood, A., Corman, J.R., Jankowski, K.J., Carey, J.C., Conley, D.J., Elmstrom, E., Fernandez, N.M., Stewart, B., Wang, S., **Wen, T.** and Sullivan, P., Anthropogenic Land Use Decouples Seasonal Nutrient Regimes and Drives Stoichiometric Variability in Rivers. Science Advances.

in review §Johnson, K., Atwood, A., Corman, J., Sethna, L., Jankowski, K.J., Carey, J., Bush, S.A., Shogren, A., Stewart, B., Conley, D., **Wen, T.**, Wang, Z., Macedo, M.N., Koenig, L. and Sullivan, P.L., Widespread non-stationarity in global river nutrient seasonality and stoichiometry. PNAS.

in review Sun, X., Planavsky, N., Hartmann, J., Amatulli, G., **Wen, T.**, Wang, Z., Li, Y., Li, S., Reinhard, C. and Zhang, S., Human land modification shifts riverine alkalinity from silicate to carbonate sources. Science Advances.

2026

52. §Fenske, A., ††Su, J., Baka, J., Forsythe, B., Williams, L., Shaheen, S.W., **Wen, T.**, Brantley, S.L. and Warner, N.R., 2026. Matrix-Derived Radium Mobilization in Shallow Aquifers Near Unconventional Oil and Gas Operations. Environmental Science & Technology. https://doi.org/10.1021/acs.est.6c05003

51. †Basijokaite, R., Kang, M. and **Wen, T.***, 2026. Large-sample evidence for surface and groundwater quality associations with marginal, orphaned, and unconventional oil and gas wells across the United States. Environmental Research Letters. https://doi.org/10.1088/1748-9326/ae7c17

50. ††Su, J., §Williams, L., §Fenske, A., §Shaheen, S., Brantley S.L., Forsythe, B., Warner, N.R., Baka, J. and **Wen, T.***, 2026. Targeted Sampling Reveals Localized Brine Signals Without Evidence of Regional Water Quality Impacts from Unconventional Oil and Gas Development. Environmental Science: Processes & Impacts. https://doi.org/10.1039/D6EM00142D

49. †††Olivier-Meehan, R., Simons, A.L., Prabhu, A., Carter, E., †Basijokaite, R., Lackey, G. and **Wen, T.***, 2026. The Legacy of Conventional Oil and Gas Development Outweighs Shale Gas Impacts on Stream Biodiversity. ACS ES&T Water. https://doi.org/10.1021/acsestwater.5c01413

48. Wang, S., Zhou, X., Huang, X., Zeng, Z., **Wen, T.**, Tian, J., He, M., Dong, J., Li, J., Yan, Y., Wang, Y., Yao, B., Xing, G., Cui, S., Yan, H., Li, R., Zheng, W., Li, L., Li, Z., Xing, L., 2026. Tectonic control on mantle-source helium migration at the southeastern Tibetan Plateau margin. Geological Society of America Bulletin. https://doi.org/10.1130/B38456.1

2025

47. Liu, R., §Xu, R., **Wen, T.***, ††Atchinson, K., Feng, Z., Hao, F., Hu, L., Tian, J., Zhang, Y., Liu, J. and Tuo, L., 2025. Noble gas constraints on fluid flow and hydrocarbon accumulation in the Yinggehai Basin, Northwestern South China Sea. Geoscience Frontiers. https://doi.org/10.1016/j.gsf.2025.102169

46. §Gunn, M., Mills, J.S., Mahoney, M., Beier, C., **Wen, T.** and Tuttle, S., 2025. A Machine Learning Approach for Estimation of Snow Depth from Snow Temperature for a Northeastern U.S. Forested Watershed. Hydrological Processes. https://doi.org/10.1002/hyp.70273

45. ††Wang, Z. and **Wen, T.***, 2025. Inferring End-members from Geoscience Data using Simplex Projected Gradient Descent-Archetypal Analysis. Journal of Geophysical Research: Machine Learning and Computation. https://doi.org/10.1029/2024JH000540

44. ††E, B., Zhang, S., Carter, E., §Meem, T.J. and **Wen, T.***, 2025. Predicting Salinity and Alkalinity Fluxes of U.S. Freshwater in a Changing Climate: Integrating Anthropogenic and Natural Influences Using Data-Driven Models. Applied Geochemistry. https://doi.org/10.1016/j.apgeochem.2025.106285

43. Liu, R., **Wen, T.***, Pinti, D.L., Hao, F., Xu, S., Shu, Z. and Xu, L., 2025. Spatiotemporal variation in tectonic deformation of the Upper Yangtze Block documented by noble gases in Paleozoic shale fluids. International Journal of Coal Geology. https://doi.org/10.1016/j.coal.2024.104671

2024

42. §Armos, B., Zhang, S., **Wen, T.**, Gellerson, E. and Daripa, P., 2024. A Harmonized River-Ocean Coupled Database for the Northern Gulf of Mexico. Scientific Data. http://doi.org/10.1038/s41597-024-04338-1

41. §Shaheen, S., **Wen, T.**, §Zheng, Z., Xue, L., Baka, J. and Brantley, S.L., 2024. Wastewaters Coproduced with Shale Gas Drive Slight Regional Salinization of Groundwater. Environmental Science & Technology. https://doi.org/10.1021/acs.est.4c03371

40. Hammond, P.A., **Wen, T.***, Woda, J. and Oakley, D., 2024. Pathways and environmental impacts of methane migration: case studies in the Marcellus Shale, USA. Geofluids. https://doi.org/10.1155/2024/9290873

39. Liu, Y., Wang, T., **Wen, T.**, Zhang, J., Liu, B., Li, Y., Zhang, H., Rong, X., Ma, L., Guo, F., Liu, X and Sun, Y., 2024. Deep learning-based grain size decomposition model: A feasible solution for dealing with methodological uncertainty. Sedimentology. https://doi.org/10.1111/sed.13195

38. ††Nesheim, S., Yu, Z., Tuttle, S., †††Klein, J. and **Wen, T.***, 2024. Assessing impacts of cemeteries on water quality in an urban headwater watershed with mixed human-built infrastructure. Hydrological Processes. https://doi.org/10.1002/hyp.15128

2023

37. Kang, M., Boutot, J., McVay, R.C., Roberts, K., Jasechko, S., Perrone, D., **Wen, T.**, Lackey, G., Raimi, D., Digiulio D.C., Shonkoff, S.B., Carey, J.W., Elliott, E.G., Vorhees, D.J. and Peltz, A., 2023. Environmental risks and opportunities of orphaned oil and gas wells in the United States. Environmental Research Letters. https://doi.org/10.1088/1748-9326/acdae7

36. §Shaughnessy, A.R., §Forgeng, M.J., **Wen, T.**, Gu, X., Hemingway, J. and Brantley, S.L., 2023.Linking Stream Chemistry to Subsurface Redox Architecture. Water Resources Research. https://doi.org/10.1029/2022WR033445

35. ††E, B., Zhang, S., Driscoll, C.T. and **Wen, T.***, 2023. Human and natural impacts on the U.S. freshwater salinization and alkalinization: A machine learning approach. Science of The Total Environment. https://doi.org/10.1016/j.scitotenv.2023.164138

34. Carter, E., Hultquist, C. and **Wen, T.**, 2023. GRRIEn Framework: a data science cheat sheet for earth scientists working at large spatial scales. Artificial Intelligence for the Earth Systems. https://doi.org/10.1175/AIES-D-22-0065.1

2022

33. §Guo, X., Liu, R., Xu, S., Feng, B., **Wen, T.**, Zhang, T., 2022. Structural deformation of shale pores in the fold-thrust belt: The Wufeng-Longmaxi shale in the Anchang Syncline of Central Yangtze Block. Advances in Geo-Energy Research. 6, 515–530. https://doi.org/10.46690/ager.2022.06.08

32. Cheng, P., Xiao, X., Ren, B., **Wen, T.** and Yu, S., 2022. Editorial: New advances in light oil/condensate geochemistry. Frontiers in Earth Science. https://doi.org/10.3389/feart.2022.1079834

31. ††Epuna, F., §Shaheen, S. and **Wen, T.***, 2022. Road Salting and Natural Brine Migration Revealed as Major Sources of Groundwater Contamination Across Regions of Northern Appalachia With and Without Unconventional Oil and Gas Development. Water Research. https://doi.org/10.1016/j.watres.2022.119128

30. **Wen, T.***, §Chen, C., Zheng, G., Bandstra, J. and Brantley, S.L., 2022. Using a Neural Network – Physics-based Hybrid Model to Predict Soil Reaction Fronts. Computers & Geosciences. https://doi.org/10.1016/j.cageo.2022.105200

29. §Shaheen, S., **Wen, T.**, Herman, A. and Brantley, S.L., 2022. Geochemical Evidence of Potential Groundwater Contamination with Human Health Risks Where Hydraulic Fracturing Overlaps with Extensive Legacy Hydrocarbon Extraction. Environmental Science & Technology. https://doi.org/10.1021/acs.est.2c00001

28. §Wang, S., Huang, X., **Wen, T.**, Wang, X., Wang, H., Han, Y., Li, Z., Kuang, J. and Qi S., 2022. Noble Gases in Shallow Aquifers Preserve Signatures of Boiling Events beneath Weishan Volcano of Wudalianchi Volcanic Field, Northeast China. Journal of Hydrology. https://doi.org/10.1016/j.jhydrol.2022.128246

27. Boumaiza, L., Walter, J., Chesnaux, R., Stotler, R., **Wen, T.**, Johannesson K., Brindha K. and Huneau, F., 2022. Chloride-salinity as indicator of the chemical composition of groundwater: empirical predictive model based on aquifers in Southern Quebec, Canada. Environmental Science and Pollution Research. https://doi.org/10.1007/s11356-022-19854-z

26. He, Y., Zhou, Y., **Wen, T.**, Zhang, S., Huang, F., Zou, X., Ma, X., Zhu, Y., 2022. A review of machine learning in geochemistry and cosmochemistry: Method improvements and applications. Applied Geochemistry. https://doi.org/10.1016/j.apgeochem.2022.105273

2021

25. Brantley, S.L., **Wen, T.**, Agarwal, D., Catalano, J.G., Schroeder, P.A., Lehnert, K., Varadharajan, C., Pett-Ridge, J., Engle, M., Castronova, A.M., Hooper, R.P., Ma, X., Jin, L., McHenry, K., Aronson, E., Shaughnessy, A.R., Derry, L.A., Richardson, J., Bales, J., Pierce, E.M., 2021. The future low-temperature geochemical data-scape as envisioned by the U.S. geochemical community. Computers & Geosciences. https://doi.org/10.1016/j.cageo.2021.104933

24. Liu, R., **Wen, T.***, Amalberti, J., Zheng, J., Hao, F. and Jiang D., 2021. The Dichotomy in Noble Gas Signatures Linked to Tectonic Deformation in Wufeng-Longmaxi Shale, Sichuan Basin. Chemical Geology. https://doi.org/10.1016/j.chemgeo.2021.120412

23. Zheng, G., Liu, C., Wei, H., Jenkins, P., Chen, C., **Wen, T.** and Li, Z., 2021, August. Knowledge-based Residual Learning. In Proceedings of the 30th International Joint Conference on Artificial Intelligence (IJCAI 2021). https://doi.org/10.24963/ijcai.2021/228

22. **Wen, T.***, Liu, M., Woda, J., Zheng, G. and Brantley, S.L., 2021. Detecting Anomalous Methane into Groundwater within Hydrocarbon Production Areas across the United States. Water Research. https://doi.org/10.1016/j.watres.2021.117236

21. §Shaughnessy, A., Gu, X., **Wen, T.** and Brantley, S.L., 2021. Machine Learning Deciphers CO2 Sequestration and Subsurface Flowpaths from Stream Chemistry. Hydrology and Earth System Sciences. https://doi.org/10.5194/hess-25-3397-2021

20. Niu, X., **Wen, T.** and Brantley, S.L., 2021. Exploring the trend of stream sulfate concentrations as U.S. power plants shift from coal to shale gas. Environmental Pollution. https://doi.org/10.1016/j.envpol.2021.117102

pre-2021

19. §Agarwal, A.*, **Wen, T.***, §Chen, A., §Zhang, A.Y., Niu, X., Zhan, X., Xue, L., Brantley, S.L., 2020. Assessing Contamination of Stream Networks Near Shale Gas Development Using a New Geospatial Tool. Environmental Science & Technology. https://doi.org/10.1021/acs.est.9b06761

18. §Woda, J., **Wen, T.**, Lemon, J., §Marcon, V., Keeports, C.M., Zelt, F., Steffy, L.Y. and Brantley, S.L., 2020. Methane concentrations in streams reveal gas leak discharges in regions of oil, gas, and coal development. Science of The Total Environment. https://doi.org/10.1016/j.scitotenv.2020.140105

17. Hammond, P.A., **Wen, T.**, Brantley, S.L. and Engelder, T., 2020. Gas well integrity and methane migration: evaluation of published evidence during shale-gas development in the USA. Hydrogeology Journal. https://doi.org/10.1007/s10040-020-02116-y

16. §Shaughnessy, A., **Wen, T.**, Niu, X. and Brantley, S.L., 2019. Three Principles to Use in Streamlining Water Quality Research through Data Uniformity. Environmental Science & Technology. https://doi.org/10.1021/acs.est.9b06406

15. **Wen, T.***, §Woda, J., §Marcon, V., Niu, X., Li, Z. and Brantley, S.L., 2019. Exploring How to Use Groundwater Chemistry to Identify Migration of Methane near Shale Gas Wells in the Appalachian Basin. Environmental Science & Technology. https://doi.org/10.1021/acs.est.9b02290

14. Liu, R., Heinemann, N., Liu, J., Zhu, W., Wilkinson, M., Xie, Y., Wang, Z., **Wen, T.**, Hao, F., Haszeldine, S.R., 2019. CO2 Sequestration by Mineral Trapping in Natural Analogues in the Yinggehai Basin, South China Sea. Marine and Petroleum Geology. https://doi.org/10.1016/j.marpetgeo.2019.03.018

13. **Wen, T.***, §Agarwal, A.*, Xue, L., §Chen, A., Herman, A., Li, Z. and Brantley, S.L., 2019. Assessing Changes in Groundwater Chemistry in Landscapes with More than 100 Years of Oil and Gas Development. Environmental Science: Processes & Impacts. http://doi.org/10.1039/C8EM00385H

12. §Woda, J., **Wen, T.**, §Oakley, D., Yoxtheimer, D., Engelder, T., Castro, M.C. and Brantley, S.L., 2018. Detecting and Explaining Why Aquifers Occasionally Become Degraded Near Hydraulically Fractured Shale Gas Wells. Proceedings of the National Academy of Sciences, 115(49), pp.12349-12358. http://doi.org/10.1073/pnas.1809013115

11. Larson, T.E., Nicot, J.P., Mickler, P., Castro, M.C., Darvari, R., **Wen, T.** and Hall, C.M., 2018. Monitoring Stray Natural Gas in Groundwater with Dissolved Nitrogen. An Example from Parker County, Texas. Water Resources Research, 54(9), pp.6024-6041. http://doi.org/10.1029/2018WR022612

10. **Wen, T.***, Niu, X., Gonzales, M., §Zheng, G., Li, Z. and Brantley, S.L., 2018. Big Groundwater Data Sets Reveal Possible Rare Contamination Amid Otherwise Improved Water Quality for Some Analytes in a Region of Marcellus Shale Development. Environmental Science & Technology, 52(12), pp.7149-7159. http://doi.org/10.1021/acs.est.8b01123

9. Niu, X., **Wen, T.**, Li, Z. and Brantley, S.L., 2018. One Step toward Developing Knowledge from Numbers in Regional Analysis of Water Quality. Environmental Science & Technology, 52(6), pp.3342-3343. http://doi.org/10.1021/acs.est.8b01035

8. **Wen, T.***, Pinti, D.L., Castro, M.C., López-Hernández, A., Hall, C.M., Shouakar-Stash, O. and Sandoval-Medina, F., 2018. A Noble Gas and 87Sr/86Sr Study in Fluids of the Los Azufres Geothermal Field, Mexico – Assessing Impact of Exploitation and Constraining Heat Sources. Chemical Geology, 483, pp.426-441. http://doi.org/10.1016/j.chemgeo.2018.03.010

7. Brantley, S.L., Vidic, R.D., Brasier, K., Yoxtheimer, D., Pollak, J., Wilderman, C. and **Wen, T.**, 2018. Engaging over data on fracking and water quality. Science, 359(6374), pp.395-397. http://doi.org/10.1126/science.aan6520

6. **Wen, T.***, Castro, M.C., Nicot, J.P., Hall, C.M., Pinti, D.L., Mickler, P., Darvari, R. and Larson, T., 2017. Characterizing the noble gas isotopic composition of the Barnett Shale and Strawn group and constraining the source of stray gas in the Trinity Aquifer, north-central Texas. Environmental Science & Technology, 51(11), pp.6533-6541. http://doi.org/10.1021/acs.est.6b06447

5. **Wen, T.***, Castro, M.C., Nicot, J.P., Hall, C.M., Larson, T., Mickler, P. and Darvari, R., 2016. Methane Sources and Migration Mechanisms in Shallow Groundwaters in Parker and Hood Counties, Texas - A Heavy Noble Gas Analysis. Environmental Science & Technology, 50(21), pp.12012-12021. http://doi.org/10.1021/acs.est.6b01494

4. **Wen, T.***, Castro, M.C., Ellis, B.R., Hall, C.M. and Lohmann, K.C., 2015. Assessing compositional variability and migration of natural gas in the Antrim Shale in the Michigan Basin using noble gas geochemistry. Chemical Geology, 417, pp.356-370. http://doi.org/10.1016/j.chemgeo.2015.10.029

3. **Wen, T.***, Castro, M.C., Hall, C.M., Pinti, D.L. and Lohmann, K.C., 2015. Constraining groundwater flow in the Glacial Drift and Saginaw aquifers in the Michigan Basin through helium concentrations and isotopic ratios. Geofluids, 16(1), pp.3-25. http://doi.org/10.1111/gfl.12133

2. Boucher, C., Pinti, D.L., Roy, M., Castro, M.C., Cloutier, V., Blanchette, D., Larocque, M., Hall, C.M., **Wen, T.** and Sano, Y., 2015. Groundwater age investigation of eskers in the Amos region, Quebec, Canada. Journal of Hydrology, 524, pp.1-14. http://doi.org/10.1016/j.jhydrol.2015.01.072

1. Nie, Y., Liu, X., **Wen, T.**, Sun, L. and Emslie, S.D., 2014. Environmental implication of nitrogen isotopic composition in ornithogenic sediments from the Ross Sea region, East Antarctica: Δ15N as a new proxy for avian influence. Chemical Geology, 363, pp.91-100. http://doi.org/10.1016/j.chemgeo.2013.10.031

Book and Book Chapters

3. **Wen, T.**, 2021. Data Mining. In: Daya Sagar, B.S., Cheng, Q., McKinley, J., Agterberg, F. (eds) Encyclopedia of Mathematical Geosciences. Springer, Cham, Switzerland. https://doi.org/10.1007/978-3-030-26050-7_77-1.

2. **Wen, T.**, 2020. Data Sharing. In: Schintler, L., McNeely, C. (eds) Encyclopedia of Big Data. Springer, Cham, Switzerland. 3pp. http://doi.org/10.1007/978-3-319-32001-4_322-1.

1. **Wen, T.**, 2020. Data Aggregation. In: Schintler, L., McNeely, C. (eds) Encyclopedia of Big Data. Springer, Cham, Switzerland. 4pp. http://doi.org/10.1007/978-3-319-32001-4_296-1.

Reports and Theses

5. Baka, J., Brantley, S.L., **Wen, T.**, Xue, L., §Shaheen, S. and §Harrington, O., 2025. Using Geoscientific Analysis and Community Engagement to Analyze Exposures to Potential Groundwater Contamination Related to Hydrocarbon Extraction in Southwestern Pennsylvania (Technical Report, https://www.heienergy.org/publication/using-geoscientific-analysis-and-community-engagement-analyze-exposures-potential).

4. **Wen, T.**, 2017. Development of Noble Gas Techniques to Fingerprint Shale Gas and to Trace Sources of Hydrocarbons in Groundwater (Doctoral dissertation, University of Michigan).

3. Nicot, JP., et al., 2015. Understanding and Managing Environmental Roadblocks to Shale Gas Development: An Analysis of Shallow Gas, NORM, and Trace Metals (Technical Report, http://www.rpsea.org/projects/11122-56/).

2. **Wen, T.**, 2014. Constraining groundwater flow in the Glacial Drift and Saginaw Aquifers in the Michigan Basin through helium concentrations and isotopic ratios (Master thesis, University of Michigan).

1. **Wen, T.**, 2011. Analysis on nitrogen species and isotopic composition of the ornithogenic sediments from Cape Bird, Ross Island, East Antarctica (Bachelor thesis, University of Science and Technology of China).

Preprints and Postprints

5. Earth Science Information Partners, & Geological Society of America. (2025). Data Help Desk @ GSA Connects 2025. The Geological Society of America Connects 2025 (GSA Connects 2025), San Antonio, Texas, USA. Zenodo. https://doi.org/10.5281/zenodo.17458265

4. Brantley, S.L., **Wen, T.**, Agarwal, D., Catalano, J., Schroeder, P.A., Lehnert, K., Varadharajan, C., Pett-Ridge, J., Engle, M., Castronova, A.M. and Hooper, R., A Vision for the Future Low-Temperature Geochemical Data-scape. EarthArXiv. https://doi.org/10.31223/X5ZP5W.

3. Shaughnessy, A.R., Gu, X., **Wen, T.** and Brantley, S.L., 2020. Machine Learning Deciphers CO2 Sequestration and Subsurface Flowpaths from Stream Chemistry. Hydrology and Earth System Sciences Discussions. https://doi.org/10.5194/hess-2020-537.

2. Woda, J., **Wen, T.**, Lemon, J., Marcon, V., Keeports, C.M., Zelt, F., Steffy, L.Y. and Brantley, S.L., 2020. Methane concentrations in streams reveal gas leak discharges in regions of oil, gas, and coal development. EarthArXiv. https://doi.org/10.31223/osf.io/qka7d.

1. Zheng, G., Liu, M., **Wen, T.**, Wang, H., Yao, H., Brantley, S.L. and Li, Z., 2019. Targeted Source Detection for Environmental Data. arXiv preprint. arXiv:1908.11056.

Open Access Datasets and Code

15. **Wen, T.**, 2025. Datasets and codes for the peer review article E et al. (2025) "Predicting Salinity and Alkalinity Fluxes of U.S. Freshwater in a Changing Climate: Integrating Anthropogenic and Natural Influences Using Data-Driven Models", HydroShare. https://doi.org/10.4211/hs.dcf1135ac217455d90f62c709cecabc6

14. **Wen, T.**, 2024. Noble gases in Paleozoic shale fluids document tectonic events and fluid migration in the Upper Yangtze Block, Zenodo. https://doi.org/10.5281/zenodo.6963863

13. **Wen, T.**, 2023. Datasets and codes for the peer review article "Human and natural impacts on the U.S. freshwater salinization and alkalinization: A machine learning approach" (1.0), Zenodo. https://doi.org/10.5281/zenodo.7937820

12. **Wen, T.**, 2022. HANDS-Research-Group/HNN_Soil_Reaction_Front: Predicting Soil Reaction Front Using Hybrid Neural Network (v1.0.0), Zenodo. https://doi.org/10.5281/zenodo.6974714

11. **Wen, T.** and Brantley, S.L., 2022. A snapshot of the entire Shale Network database as of June 2022, Data Commons, Penn State University. https://doi.org/10.26208/z1hh-bx03

10. Shaheen, S.W., **Wen, T.**, Herman, A. and Brantley, S.L., 2022. Geochemical Evidence of Potential Groundwater Contamination with Human Health Risks Where Hydraulic Fracturing Overlaps with Extensive Legacy Hydrocarbon Extraction, Data Commons, Penn State University. https://doi.org/10.26208/2nqe-wd53

9. **Wen, T.**, 2021. Sliding Window Geospatial Tool for Analyzing Geospatial Relationship in Earth and Environmental Sciences Data (v1.0), Zenodo. https://doi.org/10.5281/zenodo.5450671

8. **Wen, T.** and Brantley, S.L., 2021. Archived Dataset for Wen et al. (2021) in Water Research, Data Commons, Penn State University. https://doi.org/10.26208/qs4c-ks39

7. **Wen, T.** and Brantley, S.L., 2021. Shale Network database – A snapshot of the Entire Shale Network database as of March 2021, Data Commons, Penn State University. https://doi.org/10.26208/41rm-q220

6. **Wen, T.** and Brantley, S.L., 2021. Shale Network – Production, Injection, and Flowback Water Geochemistry in shale gas development area in the northeastern USA as of March 2021, Data Commons, Penn State University. https://doi.org/10.26208/0f3v-zr37

5. **Wen, T.**, Woda, J., Marcon, V., Gonzales, M., Niu, X., Herman, A., Guarnieri, M., Li, Z., Brantley, S.L., 2019. Shale Network – Statewide Groundwater in Pennsylvania as of January 2019, Data Commons, Penn State University. https://doi.org/10.26208/8ag3-b743

4. **Wen, T.**, Niu, X., Herman, A., Guarnieri, M., Li, Z. and Brantley, S.L., 2019. Shale Network – Statewide Surface in Pennsylvania as of February 2018, Data Commons, Penn State University. https://doi.org/10.26208/dbq0-k948

3. **Wen, T.**, Gonzales, M., Niu, X., Herman, A., Guarnieri, M., Li, Z. and Brantley, S.L., 2018. Shale Network – Mercer County Groundwater as of August 2018, Data Commons, Penn State University. https://doi.org/10.18113/D3967X

2. **Wen, T.**, Woda, J., Gonzales, M., Herman, A. and Brantley, S.L., 2018. Shale Network – Lycoming County Groundwater as of October 2018, Data Commons, Penn State University. https://doi.org/10.18113/D35M2X

1. **Wen, T.**, Gonzales, M., Niu, X., Herman, A., Guarnieri, M., Li, Z. and Brantley, S.L., 2018. Shale Network – Bradford County Groundwater as of May 2018, Data Commons, Penn State University. https://doi.org/10.26208/rj0h-qf52

Online Educational Modules and Courses

7. **Wen, T.**, 2025. EAR 488/688 – Machine Learning in Earth and Environmental Sciences. https://jaywen.com/courses/ear-400-ml-ees/

6. **Wen, T.**, 2024. EAR 419/619 – Environmental Aqueous Geochemistry. https://jaywen.com/courses/ear-419-environmental-aqueous-geochemistry/

5. **Wen, T.**, Bandaragoda, C. and Harris, L., 2020. Data Science in Earth and Environmental Sciences. https://edx.hydrolearn.org/courses/course-v1:SyracuseUniversity+EAR601+2020_Fall/about

4. **Wen, T.**, Brazil, L., Brantley, S. L., Pelepko, S. and Beattie, S., 2018a. Bromide in the Allegheny River System. https://serc.carleton.edu/hydromodules/steps/191853.html

3. **Wen, T.**, Brazil, L., Brantley, S. L., Pelepko, S. and Beattie, S., 2018b. Earthquakes in Pennsylvania. https://serc.carleton.edu/hydromodules/steps/191859.html

2. **Wen, T.**, Brazil, L., Brantley, S. L., Pelepko, S. and Beattie, S., 2018c. How Pennsylvania Disposes of Brines Safely. https://serc.carleton.edu/hydromodules/steps/191889.html

1. **Wen, T.**, Brazil, L., Brantley, S. L., Pelepko, S. and Beattie, S., 2018d. Environmental Issues Related to Brine Disposal from Oil and Gas Development in Pennsylvania. https://serc.carleton.edu/hydromodules/units/191769.html

Teaching

EAR 488/688	Machine Learning in EES, Syracuse University (Students learn to describe programming concepts, apply geospatial analysis and machine-learning techniques to real-world Earth and environmental science questions, and interpret model outputs; Every other spring).
EAR 419/619	Aqueous Geochemistry, Syracuse University (Covers the chemical composition of water through the natural hydrologic cycle and methods of hydrochemical interpretation for groundwater and pollution problems through lectures, fieldwork, and laboratory work; Every fall).
EAR 200	Intro to Data Analysis in Earth Sci., Syracuse University (Covers sample-population description and comparison, data manipulation and visualization, databases, surface contouring and modeling, basic GIS, and introductory computational modeling of geologic processes; Every spring).
EAR 205	Water and Our Environment, Syracuse University (Covers the origin, occurrence, chemistry, and hydrology of water, including climate change, contamination, water supply, and water sustainability; Every other spring).
EAR 401/601	Hydrogeology, Syracuse University (Covers the fundamentals of groundwater hydrology and hydraulics; Fall 2020; Fall 2021).
EAR 600	Foundations of Geosciences, Syracuse University (Discussion-based exploration of discoveries, mysteries, and theories across the Earth system, with exchange across departmental subdisciplines; Fall 2022).
GEOSC/ GEOG 497	Data Mining in Environ. Sci., Penn State University (Applies conventional and emerging data-analytics tools to environmental-science problems through mini-lectures and hands-on projects; Fall 2019).
GEOSC 560	Kinetics, Penn State University (Co-taught; used data-driven models to assess natural and anthropogenic controls on watershed-scale weathering rates; Spring 2019).
GEOSC 560	Interim Instructor, Penn State University (Kinetics of Geological Processes; taught fundamentals of isotope geochemistry; April 2017).

EARTH 100s Teaching Assistant, University of Michigan (Multiple introduction classes of earth sciences; September 2013 - December 2016).

EARTH 477 Teaching Assistant, University of Michigan (Hydrogeology; guided more than 50 students through hands-on labs on contaminant transport from underground storage tanks and interpretation of stratigraphic information; September 2013 - December 2016).

EARTH 408 Teaching Assistant, University of Michigan (Introduction to GIS in the Earth Sciences; taught 24 students to conduct 2D and 3D spatial analysis in ArcGIS; received teaching evaluations of 4.5-5.0 out of 5.0; September 2013 - December 2016).

Mentorship

Awards Received by Advisees

2025 Cameron O'Neill Light (Syracuse University, B.S. Student), Syracuse University SOURCE Fellowship

2025 Aamna Khan (Syracuse University, B.S. Student), Department of Energy Mickey Leland Energy Fellowship

2021 Favour Epuna (Syracuse University, M.S.), GSA Graduate Student Research Grant

Postdoctoral Scholars

2026-present Xiumei Ma, AI for Geosciences

2026-present Mahnaz Sepehrmanesh, AI for Geosciences

2026-present Mohammad Tauseef, Noble Gas Geochemistry

2024-2026 Ruta Basijokaite, Environmental Data Sciences

2024-2026 Sitangshu Chatterjee, Noble Gas Geochemistry, Now at Bhabha Atomic Research Centre, India

Graduate Students

2025-present Ryan Olivier-Meehan, M.S. Student (Earth and Environmental Sciences), Primary Advisor

2023-present Khi Atchinson, Ph.D. Student (Earth and Environmental Sciences), Primary Advisor

2023-present Jianfeng (Nick) Su, Ph.D. Student (Earth and Environmental Sciences), Primary Advisor

2022-present Zanchenling Wang, Ph.D. Student (Earth and Environmental Sciences), Primary Advisor

2021-2023 Samuel Nesheim, M.S. (Earth and Environmental Sciences), Primary Advisor, Now Environmental Scientist at the Minnesota Pollution Control Agency

2021-2023 Beibei E, M.S. (Earth and Environmental Sciences), Primary Advisor, Now PhD Student at Texas A&M University

2020-2022 Favour Epuna, M.S. (Earth and Environmental Sciences), Primary Advisor, Now Staff Scientist at Aspect Consulting LLC

2020-2021 Rohit Patil, M.S. (Information Science and Data Analytics), Research Mentor, Now Software Development Engineer at Amazon

2024-present Raven Polk, Ph.D. Student (Earth and Environmental Sciences), Committee Member

2020-present Madison Woodley, Ph.D. Candidate (Earth and Environmental Sciences), Committee Member

2024 Changcheng Pu, Ph.D. (Civil and Environmental Engineering), Committee Member, Now Postdoctoral Fellow at National Energy Technology Laboratory

2022-2024 Lee Frank-DePue, M.S. (Earth and Environmental Sciences), Committee Member, Now Physical Science Technician at U.S. Geological Survey

2021-2023 Angela Rienzo, M.S. (Earth and Environmental Sciences), Committee Member, Now Physical Scientist at U.S. Geological Survey

2021-2023 Abigail McCarthy, M.S. (Earth and Environmental Sciences), Committee Member

2019-2023 Jeffrey Wade, Ph.D. (Earth and Environmental Sciences), Committee Member, Now Postdoctoral Fellow at NASA Jet Propulsion Laboratory

2019-2021 Danielle Minnick, M.S. (SUNY ESF), External Committee Member, Now Data Analyst at New York State Department of Health

2019-2021 Alison Rickard, M.S. (SUNY ESF), External Committee Member, Now Hydrology Technician at UC Santa Barbara

Undergraduate Students

2025-present Cameron O'Neill Light (Syracuse University, B.S. Student), Research Mentor
Environmental Sensing and Cyberinfrastructure Development

2023-2025 Ryan Olivier-Meehan (Syracuse University, B.S. Student), Research Mentor,
Now M.S. Student in my research group

2023-2025 Aamna Khan (Syracuse University, B.S. Student), Research Mentor,
Now continue as a B.S. student at SU

2022-2023 Jenna Klein (Syracuse University, B.S.), Research Mentor, Now Environmental Scientist at AECOM

2022 Linda Arterburn (SUNY ESF, B.S.), Research Mentor, Now Ph.D. Student at Georgia Institute of Technology

2020-2021 Samantha Walcott (Syracuse University, B.S.), Research Mentor, Now Associate Analyst at Eversource Energy

Other Mentorship

3. Graduate student mentorship at Penn State: Sam Shaheen (geospatial analysis and machine learning); Josh Woda (isotope geochemistry); Callum Wayman (GIS); Mengqi Liu (geoscience)
2. Undergraduate thesis supervision at Penn State: Marcus Guarnieri (2018; groundwater geochemistry in Pennsylvania)
1. Undergraduate student mentorship at University of Michigan: Guolei Han (noble gas geochemistry)

Service and Professional Activities

Service to the Profession

2026-present Associate Editor, Journal of Geophysical Research: Machine Learning and Computation

2026-present Associate Editor, Geoscience Data Journal

2026-present Member, Board of Directors for CUAHSI (Consortium of Universities for the Advancement of Hydrologic Sciences, Inc)

2025-present Member, Advisory Committee for CUAHSI (Consortium of Universities for the Advancement of Hydrologic Sciences, Inc)

2021-present University Representative, CUAHSI

2023-present Associate Editor, Applied Geochemistry

2018-present Steering and Organizing Committee for Shale Network Workshop

2020-present Panelist for National Science Foundation CSSI and U.S. Department of Energy BER

2021-present Proposal reviewer for the U.S. Department of Energy BER and ASCR, CHIST-ERA of European Union, National Science Foundation (NSF), Earth Science Information Partners (ESIP)

2015-present Manuscript Peer Reviewer, 2015 – Present: Nature Communications, Geochimica et Cosmochimica Acta, Water Resources Research, Environmental Science & Technology, Water Research, Science of the Total Environment, Chemical Geology, ACS Earth and Space Chemistry, AGU GeoHealth, Journal of Environmental Informatics, Water, Geological Society of America Today, Applied Geochemistry, Environmental Science: Processes & Impacts, Current Opinion in Environmental Science & Health, Journal of Great Lakes Research, Advances in Polar Science, Geosciences, Geoscience Data Journal, Hydrogeology Journal, Applied Computing & Geosciences, Environmental Monitoring and Assessment

2018-present International and National Conference Session Convener and Workshop Organizer

26. 2026 AGU Annual Meeting session entitled “Hydrogeochemistry Across Large Samples and Scales: Datasets, Patterns, and Drivers” (H079), Primary Convener
25. 2026 Geological Society of America Connects session entitled “Machine Learning and Artificial Intelligence in Groundwater Science: Applications, Challenges, and Future Directions”, Co-convener
24. 2026 Geological Society of America Connects session entitled “From Abandonment to Opportunity: Identifying, Monitoring, Mitigating, and Repurposing Oil and Gas Wells”, Session Advocate
23. 2026 International Association for Mathematical Geosciences Annual Conference session entitled “Theories and Practices in AI for Geosciences”, Co-convener
22. 2026 International Association for Mathematical Geosciences Annual Conference Learning Workshop entitled “Fundamental Deep Learning Concepts for Applied Geoscientists”, Co-instructor
21. 2026 DINGUE IX, Co-chair of the session “Helium (and hydrogen) exploration: the new frontiers of energy transition”
20. 2025 Geological Society of America Annual Meeting session entitled “Expanding Geology’s Horizons: Geoinformatics, Open Science, and Open Data”
19. 2025 Geological Society of America Annual Meeting session entitled “A Showcase of Student Research in Geoinformatics and Data Science”
18. 2025 Geological Society of America Annual Meeting session entitled “Fixing the Silent Leak: Identifying, Quantifying, Prioritizing, and Mitigating the Environmental and Health Impacts of Legacy Oil and Gas Drilling in North America”

17. 2024 Goldschmidt session entitled “Rates and Dates: Advances in kinetics and applications in noble gas and thermo-geochronologic systems”
16. 2024 AGU Fall Meeting Town Hall entitled “Leveraging Science on Schema to Make Research Data Findable: Where Are We Now?”
15. 2024 Geological Society of America Annual Meeting session entitled “Quantifying and Addressing the Environmental and Health Impacts of Active, Orphaned, and Abandoned Oil & Gas Wells”
14. 2024 Geological Society of America Annual Meeting session entitled “Open Science, Open Data: Geoinformatics and Why it Should be on Everyone’s Radar”
13. 2023 AGU Fall Meeting Learning Workshop entitled “End Member Mixing Analysis in Hydrogeochemistry with Case Studies”
12. 2023 AGU Fall Meeting Town Hall entitled “Use Case Applications for a Democratized Cyberinfrastructure for Open Discovery to Enable Research”
11. 2023 AGU Fall Meeting session entitled “Enhanced Rock Weathering for Carbon Dioxide Removal: From Soils to the Ocean”
10. 2023 Geological Society of America Annual Meeting session entitled “No Well Left Behind: Quantifying Impacts and Improving Management of Active, Abandoned, and Orphaned Oil and Gas Wells to Safeguard Air, Water, and Land Resources”
9. 2022 Goldschmidt session entitled “Data-driven research in geochemistry”
8. 2021 Geological Society of America Annual Meeting session entitled “Machine Learning for Advancing Data Analysis Toolkit in Geoscience”
7. 2020 Goldschmidt session entitled “Application of novel gas geochemical tools to future GeoEnergy resources Crustal Fluid Geochemistry in Energy-Related Systems: Empirical, Experimental, and Modelling Advances”
6. 2020 Goldschmidt session entitled “Development of Big Data Geochemical Networks and new Analysis and Visualization tools: Innovative approaches for 21st Century Multidimensional and Transdisciplinary Science”
5. 2019 AGU Fall Meeting session entitled “Data-Driven Discoveries in Volcanology, Geochemistry and Petrology”
4. 2019 AGU Fall Meeting session entitled “Data-driven approach to understand Earth system”
3. 2018 Goldschmidt session entitled “Using Geochemistry and Big Data to Understand the Biological-Geological Co-evolution of the Critical Zone - Including Human Impacts”
2. 2018 AGU Fall Meeting session entitled “Data Science and Geochemistry: Applying a Data-driven Approach in Geochemistry-centric Studies”
1. 2018 Shale Network Workshop, Penn State University, Workshop Instructor (May; created and delivered a computer module and hands-on water-chemistry exercise about Marcellus-related spills for more than 40 participants; helped organize a mock-spill field trip)

2025-2026 Member, Organizing Committee of the DINGUE (Developments In Noble Gas Understanding and Expertise) IX Workshop

2024-2025 Member, GSA Joint Technical Program Committee (JTPC)

2024 Mentor, Geological Society of America (GSA) On To the Future Scholar Program

2025-2026 Past Chair, GSA Geoinformatics and Data Science Division

2024-2025 Chair, GSA Geoinformatics and Data Science Division

2023-2024 Vice Chair, GSA Geoinformatics and Data Science Division

2023-2025 Committee Chair, Education & Outreach Committee, CUAHSI

2021-2025	Standing Committee, Education & Outreach Committee, CUAHSI
2021-2024	Guest Editor, Frontiers in Earth Science
2022	NSF Convergence Accelerator workshop: Managing Water for a Changing Planet
2020	Steering Committee for the NSF-funded workshop: Mapping a Future for Management of Low-Temperature Geochemical Data
2018, 2023	Judge, AGU Fall Meeting Outstanding Student Presentation Award
2018-2025	Member, Hydrologic Information System User Committee, CUAHSI
2019	Coordinator, AGU Fall Meeting Outstanding Student Presentation Award
2019	Reviewer, AGU Fall Meeting student travel grant
2018	Judge for PSU Geosciences Graduate Student Colloquium
2018	Judge for 11th Annual Postdoctoral Research Exhibition at Penn State
2015-2016	Co-founder and vice-president of AAPG student chapter at University of Michigan
2013-2015	President of USTC Alumni Association in Greater Detroit area

Service at the University and College Level

2022-present	Portfolio Reviewer for the Women in Science and Engineer Future Professionals Program (WiSE-FPP) Associates
2025-2026	Search Committee, Solid Earth Faculty Search
2023	Search Committee, Computational and Field Specialist
2022	Reviewer for the Syracuse Office of Undergraduate Research & Creative Engagement (SOURCE) Program
2021-2022	Search Committee, Hydrology and Environment Faculty Search

Service at the Department Level

2024-present	Member, Graduate Admission Committee
2022-present	Faculty Advisor, Undergraduate Organization of Geosciences (UGOGeo)
2025	Member, Review Committee for the Inaugural Thonis Family Postdoctoral Research Fellowship
2023	Member, Research Excellence Doctoral Funding Program (REDF) Selection Committee
2022-2023	Member, Strategic Planning and Visioning Committee
2021-2024	Member, IT Committee
2020-2024	Member, Analytical Facilities Committee

Professional Development

2020	NSF Geobiology and Low-Temperature Geochemistry Data Workshop: Mapping a Future for Management of Low-Temperature Geochemical Data, Atlanta, GA

2019 CUAHSI DIY Water Monitoring, Data Portals, and Watershed Modeling Workshop, Stroud Water Research Center, Avondale, PA

2018 83rd Annual Field Conference of Pennsylvania Geologists: the Triassic-Jurassic rift system of eastern North America, Center Valley, PA

2018 GeoDeepDive workshop 2018, UW-Madison, Madison, WI

2018 Data Science in Geochemistry Workshop, Goldschmidt, Boston, MA

2015 Sequence Stratigraphy short course attendee, AAPG, Denver, CO

2008-2020 Field Work:

5. Groundwater, surface water, stray gas, and sediment sampling within the Marcellus Shale footprint (monthly), 2017 – 2020
4. Natural gas sampling in the Antrim Shale area (Gas & Oil Wells), MI (1 week), 2013 – 2014
3. Groundwater sampling in the Glacial Drift aquifer in Michigan Basin (3 days), 2012
2. Mountain Huangshan in Anhui, China (4 days), 2010
1. Tai Lake, Chao Lake, Yancheng National Natural Reserve, China (1 month), 2008 – 2009

Outreach and Science Communication

2026

21. Bernardi, D., 2026a. Accelerating the Search for Rare Earth Elements. Syracuse University Today. URL https://news.syr.edu/2026/08/31/accelerating-the-search-for-rare-earth-elements/ (accessed 9.21.26).
20. Bernardi, D., 2026b. A&S Earth Sciences Professor Honored as 40 Under 40 Rising Star. Syracuse University Today. URL https://news.syr.edu/2026/07/30/as-earth-sciences-professor-honored-as-40-under-40-rising-star/ (accessed 9.21.26).
19. Bernardi, D., 2026c. Conventional Oil Drilling Leaves Deeper Mark on Waterways. Syracuse University Today. URL https://news.syr.edu/2026/03/25/conventional-oil-drilling-leaves-deeper-mark-on-waterways/ (accessed 9.21.26).
18. Bernardi, D., 2026d. Bold New Strategic Plan Energizes the College of Arts and Sciences. Syracuse University Today. URL https://news.syr.edu/2026/03/05/bold-new-strategic-plan-energizes-the-college-of-arts-and-sciences/ (accessed 9.21.26).

2025

17. Lovell, D., 2025. What you should know about the debate over fluoride. Futurity. URL https://www.futurity.org/debate-over-flouride-3282192/ (accessed 5.29.25).
16. Flath-Futurity, C., 2025. How will climate change affect America's rivers? Futurity. URL https://www.futurity.org/climate-change-americas-rivers-3274322/ (accessed 4.22.25).
15. Lovell, D., 2025. Diving Deep Into the Fluoride Debate [WWW Document]. SU News. URL https://news.syr.edu/blog/2025/04/10/diving-deep-into-the-fluoride-debate/ (accessed 4.16.25).
14. Tibbetts, J.H., 2025. America's Rivers May Hold a Key to Carbon Capture [WWW Document]. College of Arts & Sciences at Syracuse University. URL https://artsandsciences.syracuse.edu/earth-sciences-department/news/americas-rivers-may-hold-a-key-to-carbon-capture/ (accessed 3.17.25).

13. Baxter, S., 2025a. Monitoring Urban Creeks in Syracuse, New York. NexSens. URL https://www.nexsens.com/case_studies/monitoring-urban-creeks-in-syracuse-new-york.htm (accessed 3.10.25).

12. Baxter, S., 2025b. Monitoring Meadowbrook Creek: Real-Time Data Collection in an Urban Creek [WWW Document]. Environmental Monitor. URL https://www.fondriest.com/news/monitoring-meadowbrook-creek-real-time-data-collection-in-an-urban-creek.htm (accessed 3.10.25).

2024

11. Wen, T., Boettiger, C., 2024. Innovations in Geo AI: Case Studies and Success Stories [WWW Document]. URL https://soundcloud.com/introductiontoopengeoai/innovations-in-geo-ai-case-studies-and-success-stories (accessed 12.13.24).

10. Bernardi, D., 2024a. Spring Disappearance and Backyard Flooding? A&S Researchers Explore if Climate Change and/or Human Intervention are to Blame [WWW Document]. College of Arts & Sciences at Syracuse University. URL https://artsandsciences.syracuse.edu/news-all/news-2024/safeguarding-a-syracuse-waterway/ (accessed 12.13.24).

9. Bernardi, D., 2024b. Creative Climate Action. Syracuse University Arts & Sciences Fall 2024 Alumni Magazine.

8. Bernardi, D., 2024c. A&S Offers More Community Learning Pathways for Arts and Sciences Undergraduates [WWW Document]. SU News. URL https://news.syr.edu/blog/2024/11/26/as-offers-more-community-learning-pathways-for-arts-and-sciences-undergraduates/ (accessed 12.13.24).

2023

7. Dzombak, R., 2023. Elderly, low-income residents most vulnerable to groundwater pollution from fracking in Appalachia [WWW Document]. AGU Newsroom. URL https://news.agu.org/press-release/elderly-low-income-residents-most-vulnerable-to-groundwater-pollution-from-fracking-in-appalachia/ (accessed 12.13.24).

6. Newcomb, T., 2023a. America's Rivers Have Secretly Gotten So Much Saltier [WWW Document]. Yahoo News. URL https://ca.news.yahoo.com/americas-rivers-secretly-gotten-much-165000410.html (accessed 12.13.24).

5. Newcomb, T., 2023b. America's Rivers Have Secretly Gotten So Much Saltier [WWW Document]. Popular Mechanics. URL https://www.popularmechanics.com/science/environment/a44579026/why-have-american-rivers-gotten-more-salty/ (accessed 12.13.24).

4. Bernardi, D., 2023. A Machine Learning Approach to Freshwater Analysis [WWW Document]. SU News. URL https://news.syr.edu/blog/2023/07/06/a-machine-learning-approach-to-freshwater-analysis/ (accessed 12.13.24).

2022

3. Staff, N., 2022. Geochemist Receives NSF Grant for Work in Developing Search Engines for Climate Change Data [WWW Document]. SU News. URL https://news.syr.edu/blog/2022/11/04/geochemist-receives-nsf-grant-for-work-in-developing-search-engines-for-climate-change-data/ (accessed 12.13.24).

2021

2. Bernardi, D., 2021a. When It Comes to the Environmental Impact of Hydrofracking vs. Conventional Gas/Oil Drilling, Research Shows the Differences May Be Minimal [WWW Document]. SU News. URL https://news.syr.edu/blog/2021/07/14/when-it-comes-to-the-environmental-impact-of-hydrofracking-vs-conventional-gas-oil-drilling-research-shows-the-differences-may-be-minimal/ (accessed 12.13.24).

1. Bernardi, D., 2021b. Hydrofracking environmental problems not that different from conventional drilling [WWW Document]. EurekAlert! URL https://www.eurekalert.org/news-releases/756808 (accessed 12.13.24).

Conference Presentations, Colloquia, and Seminars

† = Postdoc author, directly supervised by Wen at time of publication
†† = Graduate student author, directly supervised by Wen at time of publication
††† = Undergraduate student author, directly supervised by Wen at time of publication
§ = Student or postdoc author of collaborator at time of publication

2026

121. §Johnson, K., Sethna, L.R., Atwood, A., Corman, J.R., Jankowski, K.J., Carey, J.C., Conley, D.J., Elmstrom, E., Fernandez, N.M., Stewart, B., ††Wang, Z., **Wen, T.** and Sullivan, P.L., Decoupling of Nitrogen, Silica, and Phosphorus Seasonality Drives Variability in Riverine Nutrient Depletion. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted]

120. Jankowski, K.J., Bush, S.A., §Johnson, K., Sethna, L., Atwood, A., Conley, D.J., Corman, J., Elmstrom, E., Fernandez, N.M., Koenig, L., Macedo, M., Shogren, A.J., Stewart, B., ††Wang, Z., **Wen, T.**, Sullivan, P.L. and Carey, J., Long-term Changes in Riverine Silicon Relative to Nitrogen and Phosphorus are Widespread and Vary across Land Uses. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted]

119. ††Olivier-Meehan, R., ††Wang, Z., Mohammed, A. and **Wen, T.**, Archetypal Analysis Constrains the Persistence of Acid-Deposition-Derived Sulfate in Eastern U.S. Watersheds. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted]

118. Sullivan, P.L., §Johnson, K., Atwood, A., Bush, S.A., Sethna, L., Jankowski, K.J., Carey, J., Corman, J., Conley, D.J., Elmstrom, E., Koenig, L., Fernandez, N., Macedo, M., Shogren, A., Stewart, B., ††Wang, Z. and **Wen, T.**, The Fingerprints of Rivers: Emergent Patterns in Global Geochemistry. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted; invited contribution]

117. ††Wang, Z. and **Wen, T.**, Snapping Sites to Streams: AI-Assisted Geolocation of River Monitoring Sites for Hydrological Data Infrastructure. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted]

116. ††Wang, Z., Sun, X., Zhang, S., §Johnson, K., Carey, J.C., Stewart, B., Sullivan, P.L., Jankowski, K.J., Shogren, A., Atwood, A., Bush, S.A., Corman, J.R., Conley, D.J., Elmstrom, E., Fernandez, N.M., Macedo, M., Koenig Snyder, L.E., Sethna, L.R. and **Wen, T.**, Continental-Scale Patterns and Controls of Chemical Weathering Revealed by Large River Hydrogeochemical Data. AGU Annual Meeting, San Francisco, CA, 12/2026. [Accepted]

115. ††Atchinson, K., Kyaw, R., Curtice, J., Barry, P.H., Mohammed, A. and **Wen, T.**, Constraining Greenhouse Gas-Water Interactions in Thawing Arctic Permafrost Using Noble Gas Solubility Modeling. DINGUE IX, University of Ottawa, Ottawa, ON, Canada, 07/2026. [Poster]

114. §Johnson, K., Sethna, L., Atwood, A., Corman, J., Jankowski, K.J., Carey, J., Conley, D., Elmstrom, E., Fernandez, N., Macedo, M., Stewart, B., ††Wang, Z., **Wen, T.** and Sullivan, P., Anthropogenic Land Use Drives Decoupling of Nutrient Regimes. 12th BIOGEOMON Conference, Umeå, Sweden, 06/2026. [Oral]

113. ††Su, J., Mohammed, A.A. and **Wen, T.**, Three-Dimensional Contaminant Transport Modeling of a Surface Spill in Fractured Appalachian Plateaus Bedrock. Groundwater Modeling and More, Princeton, NJ, 06/2026. [Poster]

112. §Johnson, K., Sethna, L., Jankowski, K.J., Carey, J., Atwood, A., Corman, J., Conley, D.J., Elmstrom, E., Fernandez, N., Macedo, M., Shogren, A., Stewart, B., ††Wang, Z., **Wen, T.** and Sullivan, P.L., Land Use Influences Nutrient Regimes and the Timing of Nutrient Depletion in Rivers. ASLO-SIL Joint Meeting, Montreal, QC, Canada, 05/2026. [Oral]

111. Khan, A., Stribley, K., Wood, D. and **Wen, T.**, Assessing Highway Construction Impacts on Urban Headwater Stream Quality. Engaged Humanities Network Community Showcase, Salt City Market, Syracuse, NY, 05/2026. [Poster]

110. **Wen, T.**, Tracing and Predicting Fluid Processes Across Subsurface and Catchment Systems: Integrating Geochemical Tracers and Data-Driven Modeling. University of Calgary, Calgary, AB, Canada, 03/2026. [Invited Seminar]

109. **Wen, T.**, Integrating Interdisciplinary Data and Machine Learning to Illuminate Catchment Elemental Cycles within Coupled Human and Natural Systems, SUNY-ESF Water Science and Policy Seminar, Syracuse, NY, 03/2026. [Invited Seminar]

108. **Wen, T.**, Water Quality Monitoring, Data, Cyber-Infrastructure, and Community Engagement. Onondaga Creek Watershed Meeting, Syracuse, NY, 02/2026. [Oral]

107. **Wen, T.**, Illuminating Catchment Elemental Cycles within Coupled Human and Natural Systems. University of Texas at Dallas, Richardson, TX, 01/2026. [Invited Seminar]

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106. §Johnson, K., Jankowski, K.J., Carey, J., Sethna, L., Conley, D., Corman, J., Elmstrom, E., Snyder, L.K., Fernandez, N., ††Wang, Z, **Wen, T.** and Sulivan P., Global (a)synchrony in riverine nutrient dynamics. AGU Annual Meeting, New Orleans, LA, 12/2025. [Oral]

105. **Wen, T.**, Illuminating Elemental Cycles in River Catchments in Coupled Human and Natural Systems. Carnegie Science, Washington, D.C., 09/2025. [Invited Colloquium]

104. **Wen, T.**, Community-Engaged Research in Earth and Environmental Sciences. Syracuse University, Syracuse, NY, 08/2025. [Invited Seminar]

103. **Wen, T.**, Integrating Interdisciplinary Data and Machine Learning to Illuminate Catchment Elemental Cycles within Coupled Human and Natural Systems, 2025 International Professional Association of Chinese Earth Scientists (IPACES) Annual Meeting, Kunming, China, 07/2025. [Invited Colloquium]

102. **Wen, T.**, Bridging Interdisciplinary Data and Machine Learning to Advance Understanding of Catchment Elemental Cycles in Coupled Human and Natural Systems. Gordon Research Conference: Catchment Science: Interactions of Hydrology, Biology and Geochemistry, Andover, NH, 06/2025. [Invited Talk]

101. Baka, J., Brantley, S.L., Warner, N.R., **Wen, T.**, Fenske, A., ††Su, J., Williams, L., Forsythe, B., Pinto, C., Bitzer, Z. and Mahugu, E.W., An Analysis of Water Quality in the Oldest Hydrocarbon Basin in the USA. Health Effects Institute Annual Conference, Austin, TX, 05/2025. [Poster]

100. **Wen, T.**, Bridging Interdisciplinary Data and Machine Learning to Illuminate Carbon Cycles in Land-Water Systems. Centre de recherche GEOTOP, Canada, 04/2025. [Invited Talk]

99. **Wen, T.**, Assessing Environmental Impacts of Orphaned Oil and Gas Wells on Water Quality. American Association for the Advancement of Science Center for Scientific Evidence in Public Issues, 03/2025. [Invited Talk] [Virtual]

98. Baka, J., Brantley, S.L., Warner, N.R., **Wen, T.**, Fenske, A., ††Su, J., Williams, L., Forsythe, B., Pinto, C., Bitzer, Z. and Mahugu, E.W., An Analysis of Water Quality in the Oldest Hydrocarbon Basin in the USA. League of Women Voters Shale and Public Health Conference, Pittsburgh, PA, 02/2025. [Poster]

97. **Wen, T.**, Bridging Interdisciplinary Data and Machine Learning to Illuminate Carbon Cycles in Land-Water Systems. Eastern Institute of Technology, Ningbo, China, 02/2025. [Invited Talk] [Virtual]

2024

96. ††Wang, Z. and **Wen, T.**, A Data-driven End-Member Mixing Analysis Tool Inspired by Convex Geometry for Geoscience. AGU Annual Meeting, Washington, D.C., 12/2024. [Oral]

95. ††Qin, L., Zhang, S. and **Wen, T.**, Predicting Riverine Sulfate Fluxes Across the United States Using Machine Learning. AGU Annual Meeting, Washington, D.C., 12/2024. [Poster]

94. §Sun, X., ††Wang, Z., Suhrhoff, T., Kemeny, P., Castronova, A., **Wen, T.**, and Zhang, S., Evaluating Inverse Modeling Approaches for Rock Weathering Analysis. AGU Annual Meeting, Washington, D.C., 12/2024. [Poster]

93. †††Olivier-Meehan, R., Prabhu, A. and **Wen, T.**, Using Network Analysis of Benthic Macroinvertebrates to Unravel the Effect of Oil and Gas Production on River Ecosystem Health. AGU Annual Meeting, Washington, D.C., 12/2024. [Poster]

92. †Basijokaite, R., Lackey, G., Kang, M. and **Wen, T.**, Exploring Relationships Between Unconventional, Marginal, Orphaned Oil and Gas Wells and Water Quality in United States. AGU Annual Meeting, Washington, D.C., 12/2024. [Poster]

91. **Wen, T.**, E, B., Zhang, S., Driscoll, C. and Carter, E., Quantifying continental-scale silicate weathering fluxes and coupled human-natural impacts using machine learning techniques. Goldschmidt, Chicago, IL, 08/2024. [Poster]

90. Shaheen, S., **Wen, T.**, Roden, E. and Brantley, S., Understanding methane reactivity in the critical zone using abandoned oil and gas wells. Goldschmidt, Chicago, IL, 08/2024. [Oral]

89. **Wen, T.**, From Fossil Fuel Industry to the Global Carbon Cycle: A Multidisciplinary Approach to Assess the Human-Carbon-Climate Interactions. Syracuse University, Syracuse, NY, 04/2024. [Invited Seminar]

88. Baka, J., §Shaheen, S., **Wen, T.**, §Harrington, O., Xue, L. and Brantley, S.L., Unconventional Oil/Gas Development, Water Quality, and Environmental Hazards in the Oldest Hydrocarbon Basin in the US. Health Effects Institute Annual Conference, Philadelphia, PA, 04/2024. [Poster]

87. **Wen, T.**, A Multidisciplinary Approach to Assess the Interplay of Human, Carbon, and Climate Dynamics in Watersheds Facing Salinization. Toronto Metropolitan University, Toronto, Canada, 04/2024. [Invited Colloquium] [Virtual]

86. **Wen, T.**, Multidisciplinary Insights into Human, Carbon, and Climate Dynamics Across Natural and Altered Systems. University at Albany, Albany, NY, 03/2024. [Invited Colloquium]

85. **Wen, T.**, Multidisciplinary Insights into Human, Carbon, and Climate Dynamics Across Natural and Altered Systems. Queen's University, Kingston, Canada, 03/2024. [Invited Colloquium]

84. **Wen, T.**, Multidisciplinary Insights into Human, Carbon, and Climate Dynamics Across Natural and Altered Systems. International Professional Association of Chinese Earth Scientists, 03/2024. [Invited Seminar] [Virtual]

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83. **Wen, T.**, Liu, R., Pinti, D.L. Hao, F., Xu, S., Shu, Z. and Xu, L., Noble Gas Geochemistry Documenting Spatiotemporal Variation in Tectonic Deformation of the Upper Yangtze Block. AGU Fall Meeting, San Francisco, CA, 12/2023. [Poster]

82. Kang, M., Boutot, J., McVay, R.C., Roberts, K., Jasechko, S., Perrone, D., **Wen, T.**, Lackey, G., Raimi, D., Digiulio D.C., Shonkoff, S.B., Carey, J.W., Elliott, E.G., Vorhees, D.J. and Peltz, A., Environmental risks and opportunities of orphaned oil and gas wells in the United States. AGU Fall Meeting, San Francisco, CA, 12/2023. [Poster]

81. ††Nesheim, S., Yu, Z., Tuttle, S. and **Wen, T.**, Evaluating Impacts of A Cemetery on Urban Water Quality in A Headwater Watershed with Various Human-Built Infrastructure. AGU Fall Meeting, San Francisco, CA, 12/2023. [Poster]

80. **Wen, T.**, ††E, B., Zhang, S. and Carter E., Data-Driven Assessment of Human and Natural Impacts on Future Alkalinity Fluxes of U.S. Freshwater under Changing Climate. AGU Fall Meeting, San Francisco, CA, 12/2023. [Poster]

79. Shaheen, S., **Wen, T.**, Baka, J. and Brantley, S., Wastewater spills during shale gas development as a driver of increased groundwater salinity. GSA Annual Meeting, Pittsburgh, PA, 10/2023. [Oral]

78. **Wen, T.**, Kang, M., Shaheen, S., Brantley, S. and Baka, J., Groundwater quality assessment around orphaned and unconventional oil and gas wells in the United States. GSA Annual Meeting, Pittsburgh, PA, 10/2023. [Oral]

77. Harrington, O., Shaheen, S., **Wen, T.**, Brantley, S. and Baka, J., Something in the water? Analyzing community and geoscientific knowledge of the energy-water-health nexus in southwestern Pennsylvania. GSA Annual Meeting, Pittsburgh, PA, 10/2023. [Oral]

76. Kang, M., Boutot, J., McVay, R.C., Roberts, K., Jasechko, S., Perrone, D., **Wen, T.**, Lackey, G., Raimi, D., Digiulio D.C., Shonkoff, S.B., Carey, J.W., Elliott, E.G., Vorhees, D.J. and Peltz, A., Environmental risks and opportunities of orphaned oil and gas wells in the United States. GSA Annual Meeting, Pittsburgh, PA, 10/2023. [Oral]

75. **Wen, T.**, Exploring Human-Carbon-Climate Interactions: A Multidisciplinary Approach Bridging Fossil Fuels and the Global Carbon Cycle. Michigan State University, East Lansing, MI, 10/2023. [Invited Colloquium]

74. **Wen, T.**, Exploring Human-Carbon-Climate Interactions: A Multidisciplinary Approach Bridging Fossil Fuels and the Global Carbon Cycle. USGS New York Water Science Center, Troy, NY, 10/2023. [Invited Seminar]

73. §Shaheen, S., **Wen, T.**, Lloyd, M., House, C., Roden, E. and Brantley, S.L., Biogeochemical aquifer changes driven by methane migration from leaking oil and gas wells: observations from laboratory to field. Goldschmidt, Lyon, France, 07/2023. [Oral]

72. **Wen, T.**, A Data-driven Perspective on Carbon Fluxes from Land Water Systems. Gordon Research Conference: Catchment Science: Interactions of Hydrology, Biology and Geochemistry, Andover, NH, 06/2023. [Poster]

71. §Shaheen, S., **Wen, T.**, §Harrington, O., Xue, L., Brantley, S.L. and Baka, J., Using Geoscientific Analysis and Community Engagement to Analyze Exposure to Potential Groundwater Contamination Related to Hydrocarbon Extraction in Southwestern Pennsylvania. Health Effects Institute Annual Conference, Boston, MA, 05/2023. [Poster]

70. **Wen, T.**, From Fossil Fuel Industry to the Global Carbon Cycle: A Multidisciplinary Approach to Assess the Human-Carbon-Climate Interactions. University of Arizona, Tucson, AZ, 04/2023. [Invited Colloquium]

69. **Wen, T.**, From Fossil Fuel Industry to the Global Carbon Cycle: A Multidisciplinary Approach to Assess the Human-Carbon-Climate Interactions. University of California Riverside, Riverside, CA, 03/2023. [Invited Colloquium]

68. ††Nesheim, S., †††Klein, J., Zhi, W. and **Wen, T.**, Exploring the impacts of cemeteries on urban surface water quality in Syracuse, New York. GSA Northeastern Section Meeting, Reston, VA, 03/2023. [Poster]

67. ††E, B., Zhang, S. and **Wen, T.**, Developing machine learning models to assess anthropogenic impacts on U.S. freshwater salinization syndrome. GSA Northeastern Section Meeting, Reston, VA, 03/2023. [Oral]

66. †††Klein, J., ††Nesheim, S. and **Wen, T.**, Evaluating impacts of cemeteries on surface water quality in the Finger Lakes watershed. GSA Northeastern Section Meeting, Reston, VA, 03/2023. [Poster]

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65. Brantley, S.L., §Shaheen, S., **Wen, T.** and Bake, J., Water Quality Impacts from Shale-gas Development: from Case Study to Statistical Analysis and from Pennsylvania to the USA. AGU Fall Meeting, Chicago, IL, 12/2022. [Poster]

64. §Shaheen, S., **Wen, T.**, Sharifironizi, M., House, C., Roden, E. and Brantley, S.L., Water Quality Impacts and Hydrogeologic Methane Migration Pathways from Aging Oil and Gas Wells. AGU Fall Meeting, Chicago, IL, 12/2022. [Poster]

63. **Wen, T.**, Using Machine Learning Approaches to Address Carbon Cycling-Related Questions, Central New York Association of Professional Geologists, Syracuse, NY, 12/2022. [Invited Seminar]

62. Shaheen, S, **Wen, T.** and Brantley S.L., Increased Brine Salt Concentrations in Pennsylvania Groundwater Indicate Potential Wastewater Impacts During Unconventional Oil & Gas Development. SETAC North America 43rd Annual Meeting, Pittsburgh, PA, 11/2022. [Oral]

61. **Wen, T.**, Addressing Carbon Cycling-related Questions Using Machine Learning Approaches, University of Science and Technology of China, Hefei, China, 10/2022. [Invited Seminar] [Virtual]

60. **Wen, T.**, Addressing Water and Soil Geochemistry Questions Using Data-driven Approaches. GSA Annual Meeting, Denver, CO, 10/2022. [Invited Talk]

59. Zhang, S. and **Wen, T.**, Mapping Silicate Weathering Flux with Coupled Machine Learning Techniques. GSA Annual Meeting, Denver, CO, 10/2022. [Oral]

58. **Wen, T.**, The Shale Network and Soil Profile Chemistry Databases. Pre-GSA Workshop on Geoinformatics and Data Lab Management, Denver, CO, 10/2022. [Oral]

57. **Wen, T.**, A High-throughput Cloud-based Open-Source Platform for Monitoring Stream Water Quality. Earth Science Information Partners, Pittsburgh, PA, 07/2022. [Poster]

56. **Wen, T.**, §Chen, C., Bandstra, J. and Brantley, S.L., Developing a Physics-Informed Neural Network to Predict Soil Reaction Fronts. Goldschmidt, Honolulu, HI, 07/2022. [Poster]

55. Liu, R., **Wen, T.**, Xu, L. and Hao, F., Extensive tectonic-triggered hydrocarbon expulsion from the Paleozoic shale in the Yangtze Block, South China: Insights from noble gas geochemistry. Goldschmidt, Honolulu, HI, 07/2022. [Poster]

54. Carter, E., Hultquist, C. and **Wen, T.**, GRRIEn Framework: a data science cheat sheet for earth scientists working at large spatial scales. Frontiers in Hydrology Meeting, San Juan, Puerto Rico, 06/2022 [Oral]

53. ††Epuna, F. and **Wen, T.**, Assessing Impact of Unconventional Oil and Gas Development on Groundwater Quality in the Northern Appalachian Region with Mixed Land Uses. Shale Network Workshop, University Park, PA, 05/2022. [Poster]

52. ††Epuna, F. and **Wen, T.**, Evaluating the Impact of Unconventional Oil and Gas Drilling on Groundwater Quality in the Appalachian Basin With Mixed Land Uses. GSA Northeastern Section Meeting, Lancaster, PA, 03/2022. [Oral]

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51. §Shaheen, S., Sharifironizi, M., **Wen, T.** and Brantley, S.L., Anaerobic Oxidation of Methane Influences the Mobility of Redox-Active Contaminants in Aquifers Impacted by Leaking Oil and Gas Wells. AGU Fall Meeting, New Orleans, LA, 12/2021.

50. **Wen, T.**, §Chen, C., Zheng, G., Bandstra, J. and Brantley, S.L., Using a Physics-Based Model Guided Neural Network to Predict Soil Reaction Fronts. GSA Annual Meeting, Portland, OR, 10/2021. [Virtual]

49. **Wen, T.**, Tools for Data Management nd Data Synthesizing in Hydrogeochemistry and Watershed Science Research. CUAHSI Cyber Seminar, 10/2021. [Invited Talk] [Virtual]

48. **Wen, T.**, Shale Network Database: Data Bring Energy Stakeholders Together. CUAHSI Biennial Colloquium, 07/2021. [Virtual]

47. Liu, R., **Wen, T.**, Jiang, D., Hao, F., Han, G. and Castro M.C. Co-evolution of tectonic deformation and noble gas geochemistry in shale: an inside-out view of the Sichuan Basin. Developments in Noble Gas Understanding and Expertise VII, 07/2021. [Virtual]

46. §Shaughnessy, A., **Wen, T.** and Brantley S.L., Water Flowpath and Source Chemistry Control C-Q Relationships Across Spatial Scales. Goldschmidt, Lyon, France, 07/2021. [Virtual]

45. **Wen, T.**, Detecting Hydrocarbon Production Related Water Impairments: A Data-driven Approach Using Large Water Quality Datasets. NETL Well Integrity Workshop, Pittsburgh, PA, 05/2021. [Invited Seminar] [Virtual]

44. **Wen, T.**, Niu, X. and Brantley, S.L., Stream sulfate concentrations decrease as U.S. power plants shift from coal to shale gas. Shale Network Workshop, University Park, PA, 05/2021. [Poster]

Prior to 2021

43. §Shaughnessy, A.R., §Forgeng, M., Xin, G., **Wen, T.**, §Shaheen, S. and Brantley, S.L., Water Flowpath and Bedrock Geology Control Pyrite Weathering Across Spatiotemporal Scales. AGU Fall Meeting, San Francisco, CA, 12/2020. [Virtual]

42. Brantley, S.L., **Wen, T.**, §Shaheen, S. and §Shaughnessy, A.R., Exploring Societal Problems with Data: Assessing Impacts on Water Quality with the Shale Network Database. AGU Fall Meeting, San Francisco, CA, 12/2020. [Virtual]

41. §Shaheen, S., **Wen, T.**, Herman, A. and Brantley, S.L., Investigating the sources and extent of groundwater contamination in areas of extensive oil, gas, and coal extraction using data mining. AGU Fall Meeting, San Francisco, CA, 12/2020. [Virtual]

40. **Wen, T.**, Using Small Data and Big Data to Assess the Impact of Shale Gas Drilling on Water Quality. Zhejiang University, Hangzhou, China, 09/2020. [Invited Seminar] [Virtual]

39. Liu, R., **Wen, T.**, Zheng, J. and Hao, F., Noble Gas Geochemistry in the Wufeng-Longmaxi Shale of the Southern Sichuan Basin, China. Goldschmidt, Honolulu, HI, 06/2020. [Virtual]

38. **Wen, T.**, §Liu, M., Li, Z. and Brantley, S.L., Using Big Groundwater Data to Detect Methane Contamination in Water within Hydrocarbon Production Areas Across the United States. Goldschmidt, Honolulu, HI, 06/2020. [Virtual]

37. **Wen, T.**, §Liu, M., Li, Z. and Brantley, S.L., A machine learning-based ensemble model to detect methane contamination in groundwater within hydrocarbon production areas across the United States. AGU Fall Meeting, San Francisco, CA, 12/2019. [Poster]

36. **Wen, T.**, Niu, X, §Shaughnessy, A.R. and Brantley, S.L., Ensuring reusability of water quality data: what have we learned as both data users and providers? AGU Fall Meeting, San Francisco, CA, 12/2019. [Invited Talk]

35. §Agarwal, A., **Wen, T.**, §Chen, A., Xue, L. and Brantley, S.L., GeoNet: An automated geochemical network analysis with application to detecting stream water contamination. Annual Conference of the International Association for Mathematical Geosciences, State College, PA, 08/2019. [Oral]

34. **Wen, T.**, Using Small Data and Big Data to Assess the Impact of Shale Gas Drilling on Water Quality. Kansa Geological Survey of the University of Kansas, Lawrence, KS, 04/2019. [Invited Colloquium]

33. **Wen, T.**, Using Small Data and Big Data to Assess the Impact of Shale Gas Drilling on Water Quality. Syracuse University, Syracuse, NY, 04/2019. [Invited Colloquium]

32. **Wen, T.**, Using Small Data and Big Data to Assess the Impact of Shale Gas Drilling on Water Quality. Saint Francis University, Loretto, PA, 03/2019. [Invited Seminar]

31. **Wen, T.**, Liu, M., Woda, J., Zheng, G., Niu, X., Gonzales, M., Hall, C., Nicot, J.-P., Castro., M.C., Li, Z. and Brantley, S.L., Using Big Data and Small Data (Noble Gases) to Assess the Impact of Shale Gas Drilling on Water Quality. National Groundwater Association Workshop: Groundwater and Oil and Gas Development, San Antonio, TX, 03/2019. [Oral]

30. **Wen, T.**, §Liu, M., §Woda, J., §Zheng, G., Li, Z. and Brantley, S.L., Detecting anomalous methane in groundwater in shale gas production areas using big data. AGU Fall Meeting, Washington, D.C., 12/2018. [Poster]

29. Brantley, S.L., **Wen, T.**, Li, Z., §Liu, M., §Zheng, G., Herman, A., Gonzales, M., §Woda, J. and Niu, X., Using Big Data (and Little Data) to Understand the Effects of Shale Gas Development on Water Quality. AGU Fall Meeting, Washington, D.C., 12/2018. [Invited Talk]

28. §Woda, J., **Wen, T.**, Lemon, J., Keeports, C., Zelt, F.B. and Brantley, S.L., Using citizen science and stream methane to locate and understand hydrocarbon-related contaminant sources in Pennsylvania. AGU Fall Meeting, Washington, D.C., 12/2018. [Oral]

27. **Wen, T.**, §Zheng, G., Niu, X., §Liu, M., Li, Z. and Brantley, S.L., Using Geochemistry Data to Identify Groundwater Quality Issues in Shale Gas Production Area. Health Effects Institute Energy Research Program Workshop, Austin, TX, 09/2018. [Invited Talk]

26. **Wen, T.**, §Liu, M., §Zheng, G., Brantley, S.L. and Li, Z., Using Machine Learning to Detect Anomalous Methane in Groundwater within Shale Gas Production Areas. Goldschmidt, Boston, MA, 08/2018. [Poster]

25. Brantley, S.L., **Wen, T.**, Niu, X., §Zheng, G., Gonzales, M. and Li, Z., Using Big Groundwater Data to Understand Regional Water Chemistry. Goldschmidt, Boston, MA, 08/2018. [Poster]

24. §Woda, J., **Wen, T.** and Brantley, S.L., Distinguishing Recent Methane Migration into Groundwater from Natural Methane Sources in the Marcellus Gas Play. Goldschmidt, Boston, MA, 08/2018. [Oral]

23. **Wen, T.**, Niu, X., Pollak, J., Brazil, L., Li, Z. and Brantley, S.L., Using Shale Network Database to Assess the Water Quality Data in Marcellus Shale Area. UCOWR-NIWR Annual Water Resources Conference, Pittsburgh, PA, 06/2018. [Invited Talk]

22. **Wen, T.**, A Multi-disciplinary and Multi-stakeholder Framework to Evaluate Environmental Impacts of Shale Gas Production. Energy Days Conference, University Park, PA, 05/2018. [Oral]

21. **Wen, T.**, §Liu, M., §Zheng, G., Niu, X., Gonzales, M., §Woda, J., Li, Z. and Brantley, S.L., Applying machine learning in water quality data: implication for controlling factors and occurrence time of elevated methane in groundwater. Shale Network Workshop, University Park, PA, 05/2018. [Poster]

20. **Wen, T.**, §Zheng, G., §Liu, M., Niu, X., Gonzales, M., §Woda, J., Li, Z. and Brantley, S.L., Applying Machine Learning to Detect Anomalous Methane in Groundwater. PA Groundwater Symposium, State College, PA, 05/2018. [Oral]

19. **Wen, T.**, Niu, X., Gonzales, M., Li, Z. and Brantley, S.L., Applying Data Mining Techniques to Chemical Analyses of Pre-drill Groundwater Samples within the Marcellus Formation Shale Play in Bradford County, Pennsylvania. AGU Fall Meeting, New Orleans, LA, 12/2017. [Poster]

18. Pinti, D.L., **Wen, T.**, Castro, M.C., López-Hernández, A., Hall, C.M., Shouakar-Stash, O. and Sandoval-Medina, Using noble gases and 87Sr/86Sr to constrain heat sources and fluid evolution at the Los Azufres Geothermal Field, Mexico. AGU Fall Meeting, New Orleans, LA, 12/2017. [Poster]

17. **Wen, T.**, Castro, M.C., Nicot, J.P., Hall, C.M., Pinti, D.L., Mickler, P., Darvari, R. and Larson, T., Barnett Shale or Strawn Group: Identifying the Source of Stray Gas through Noble Gases in the Trinity Aquifer, North-Central Texas. AGU Fall Meeting, New Orleans, LA, 12/2017. [Oral]

16. **Wen, T.**, Niu, X., Gonzales, M., Li, Z. and Brantley, S.L., Using Data Mining Techniques to Assess Water Quality within the Marcellus Shale Play. Geochemistry Forum, State College, PA, 12/2017. [Oral]

15. Brantley, S.L., Gonzales, M., Guarnieri, M., Niu, X., **Wen, T.** and Li, Z., Investigating Chemical Analyses of Ground Waters Sampled by Shale-gas Industry Consultants Before Gas-well Drilling in Pennsylvania. Pennsylvania Groundwater Symposium, State College, PA, 05/2017. [Oral]

14. **Wen, T.**, Castro, M.C., Nicot, J.P., Hall, C.M., Larson, T., Mickler, P. and Darvari, R., Methane Sources and Migration Mechanisms in the Shallow Trinity Aquifer in Parker and Hood Counties, Texas – a Noble Gas Analysis. Shale Network Workshop, University Park, PA, 05/2017. [Poster]

13. **Wen, T.**, Pinti, D.L., Castro, M.C., Hall, C.M., Shouakar-Stash, O. and López-Hernández, A., Fluids in the Los Azufres Geothermal Field, Mexico traced by noble gas isotopes and 87Sr/86Sr. GAC-MAC Meeting, Kingston, Canada, 05/2017. [Poster]

12. **Wen, T.**, Development of noble gas techniques to fingerprint shale gas and to trace hydrocarbons in groundwater. China University of Geosciences, Wuhan, China, 03/2017. [Invited, Oral]

11. **Wen, T.**, Castro, M.C., Nicot, J.P., Hall, C.M., Mickler, P. and Darvari, R., Methane Sources and Migration Mechanisms in the Shallow Trinity Aquifer in Parker and Hood Counties, Texas – a Noble Gas Analysis. AGU Fall Meeting, San Francisco, CA, 12/2016. [Oral]

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